

ICT22642 – INTERNET OF THINGS (IOT)

IoT Security and Privacy - 1

S. Aysha Asra

Lecturer

Department of Information Technology
Faculty of Humanities and Social Sciences
University of Ruhuna

Introduction to IoT Security and Privacy

- The Internet of Things (IoT) connects billions of devices like smartwatches, home security systems, and industrial machines to the internet.
- These devices collect, process, and share data to make life easier and improve efficiency.
- However, as IoT grows, so do the risks associated with its use.

What is IoT Security?

- IoT security refers to the measures and practices used to protect IoT devices, networks, and the data they generate from unauthorized access, theft, or damage.
- **Key Objective:** Ensure the confidentiality, integrity, and availability of IoT systems.
- **Why Important?** IoT devices often operate in critical areas like healthcare, transportation, and homes, where a security breach can cause significant harm.

What is IoT Privacy?

Internet of Things Applications

- IoT privacy focuses on protecting personal and sensitive information collected by IoT devices. It ensures that users have control over how their data is collected, stored, and shared.
- Key Objective: Maintain user trust by safeguarding their data.
- Why Important? Many IoT devices collect highly personal information, such as health stats, location, or daily habits.

Why IoT Security and Privacy Are Critical

- **Increased Connectivity**

- With billions of devices connected, a single vulnerability can affect multiple systems.
- Example: A weakly secured smart bulb can become an entry point for hackers to access a home network.

Internet of Things Applications

- **Diverse Devices and Standards**

- IoT devices come from various manufacturers with different security standards, creating inconsistencies.
- Example: Some devices might not support encryption, making them easier targets.

Internet of Things Applications

- **Sensitive Data**

- IoT collects a large amount of data, often about personal or business activities.
- Example: Fitness trackers log health metrics, which can be exploited if leaked.

Internet of Things Applications

- **Critical Infrastructure Dependency**
- IoT is integral to essential services like energy grids, water supply, and healthcare.
- Example: A hacked smart grid could cause power outages in a city.

Challenges in IoT Security and Privacy

- Limited Device Resources
- Many IoT devices are small and lack the processing power for advanced security features like encryption.
- Example: A cheap smart sensor might not support regular updates or complex security protocols.

Internet of Things Applications

- Lack of User Awareness
- Users often do not understand the security risks of IoT devices, leading to weak passwords or unprotected networks.
- Example: A user may connect their smart TV to a public Wi-Fi network without knowing the risks.